

Every route, ranked

fill and drain, with and without a bottom port

Route	Fill	Drain	Cuts into
1 Stock pour + invert	20.0 s	15.0 s	none
2 Stock + 30° tilt technique	10.0 s	15.0 s	none
3 Funnel skirt port 1/2 in	2.7 s	15.0 s	funnel
4 Centre-column standpipe	2.5 s	9.7 s	column
5 Bottom bulkhead 1/2 in	2.7 s	10.3 s	TANK
6 Bottom bulkhead 3/4 in	1.5 s	5.9 s	TANK
7 Route 4 or 5 + recirculating pump	as above	as above	as above
8 Motorised tilt cradle	10.0 s	12.0 s	none

ANSWERS TO THE FOUR QUESTIONS

WITH A BOTTOM VALVE
Gravity fill from a raised reservoir, gravity drain to a bucket below the bench. Do not pump either direction. 1/2 in, not 3/4 in. Sheets 9 and 10.

WITHOUT A BOTTOM VALVE
Centre-column standpipe (route 4) if you will print a part; 30° tilt pouring (route 2) if you will not. Sheets 8 and 4.

PLUMBING IN AND OUT INSIDE THE FUNNEL
Two ways, and they are not equivalent. A port through the funnel SKIRT is a fast top fill and no drain at all. A tube down the CENTRE COLUMN is a fast bottom fill and a fast drain. Take the second. Sheets 7 and 8.

TILTING
Fill: 25–30° from vertical, and it keeps working until the tank is 93 % full. Worth about 2×. Drain: the geometry demands 83° to empty and full inversion to clear the reels. Tilt is a filling tool, not a draining one. Sheets 4–6.

THE RESULT THAT MATTERS

Route 4 – a printed replacement centre column with an ø18 mm through bore, open at the foot – gives 2.5 s fill and 9.7 s drain. That is bulkhead performance, from the bottom of the tank, with ZERO holes in the tank. The column is a cheap consumable spare; the tank is not.

If you were going to drill the tank for a 1/2 in port, print a column instead. It is faster, it is reversible, it cannot leak onto your bench, and it is light-tight almost by accident – the feed runs down a black tube inside a black tube.

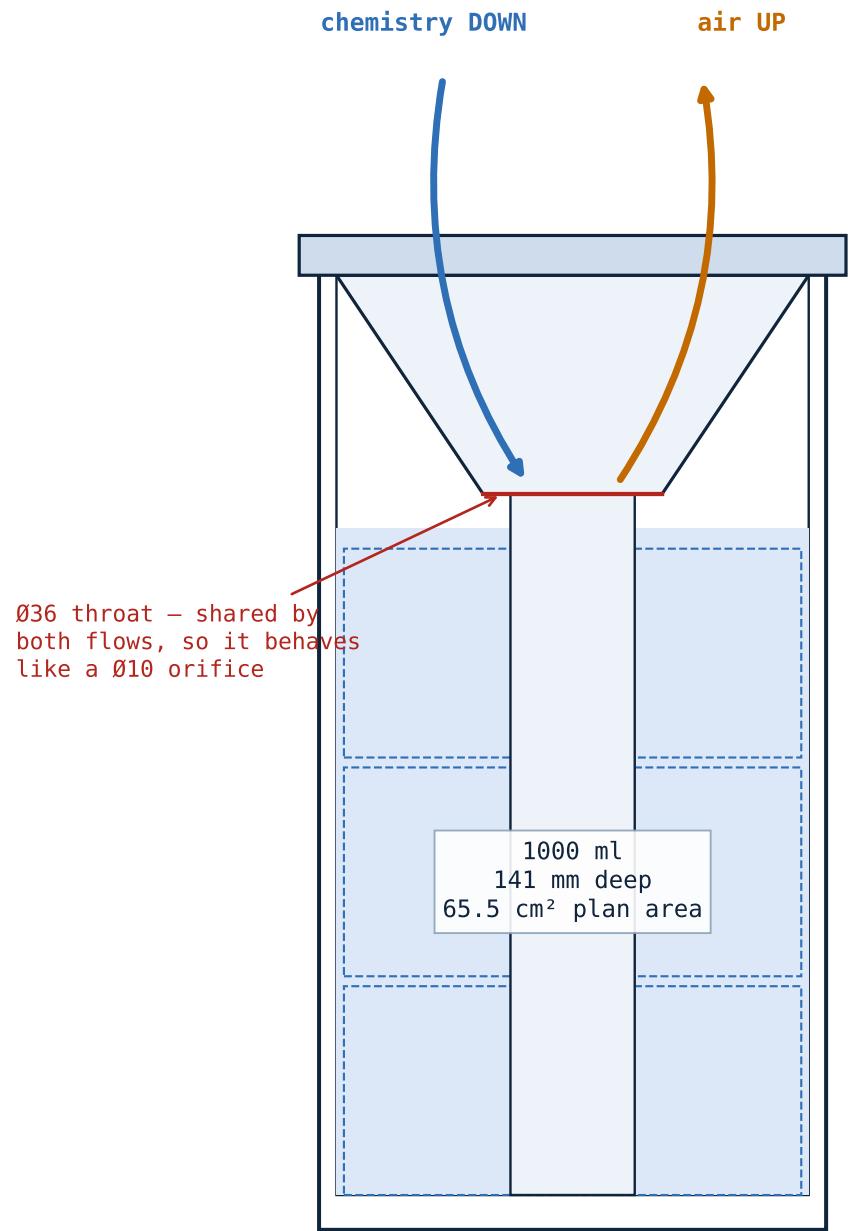
AND THE ONE THAT SURPRISES PEOPLE

Gravity beats every affordable pump. 0.6 m of head through a 1/2 in line delivers 370 ml/s – 22 litres a minute. A cheap diaphragm pump manages 4–7 L/min; a small peristaltic manages 0.5–3, slower than the stock funnel you are trying to escape. You would need a 22 L/min chemical-proof pump to match a shelf 600 mm above the bench.

Pumps are not for filling. They are for recirculating – 4 L/min is ample for continuous agitation – and for pulling the last residue out of the reels.

Baseline — why the standard tank is slow

one opening, two fluids, opposite directions



THE MEASUREMENT EVERYTHING IS ANCHORED TO

A Multi-Reel 3 takes 15–25 s to accept 1000 ml through the funnel. Call it 20 s → 50 ml/s. Back out the effective orifice:

head in the funnel	$h \approx 0.05 \text{ m}$
ideal velocity	$\sqrt{2gh} = 0.99 \text{ m/s}$
with Cd 0.6	$v = 0.60 \text{ m/s}$
area needed	$A = Q/v = 83 \text{ mm}^2$
equivalent bore	$\varnothing 10.3 \text{ mm}$

A Ø36 throat performing like a Ø10 hole. The missing 92 % of the area is occupied by air trying to get out. That is the entire problem, and every route in this document is a different way of giving the air its own way out.

COUNTER-CURRENT FLOODING — THE GOVERNING PHYSICS

Liquid down and gas up in one vertical tube is a classic flooding problem. The Wallis criterion,

$$\sqrt{j^*_g} + m \cdot \sqrt{j^*_l} = C$$
$$j^*_k = j_k \cdot \sqrt{(\rho_k / (g D \Delta \rho))}$$

says the tube jams once the two superficial velocities together exceed a constant. Below the limit you get smooth stratified flow; above it, slug flow – the glugging you can hear. Two levers fall straight out of it:

- raise D → a bigger path (routes 3–6)
- incline the tube → gravity separates the phases instead of making them fight

WHEN THIS ACTUALLY MATTERS

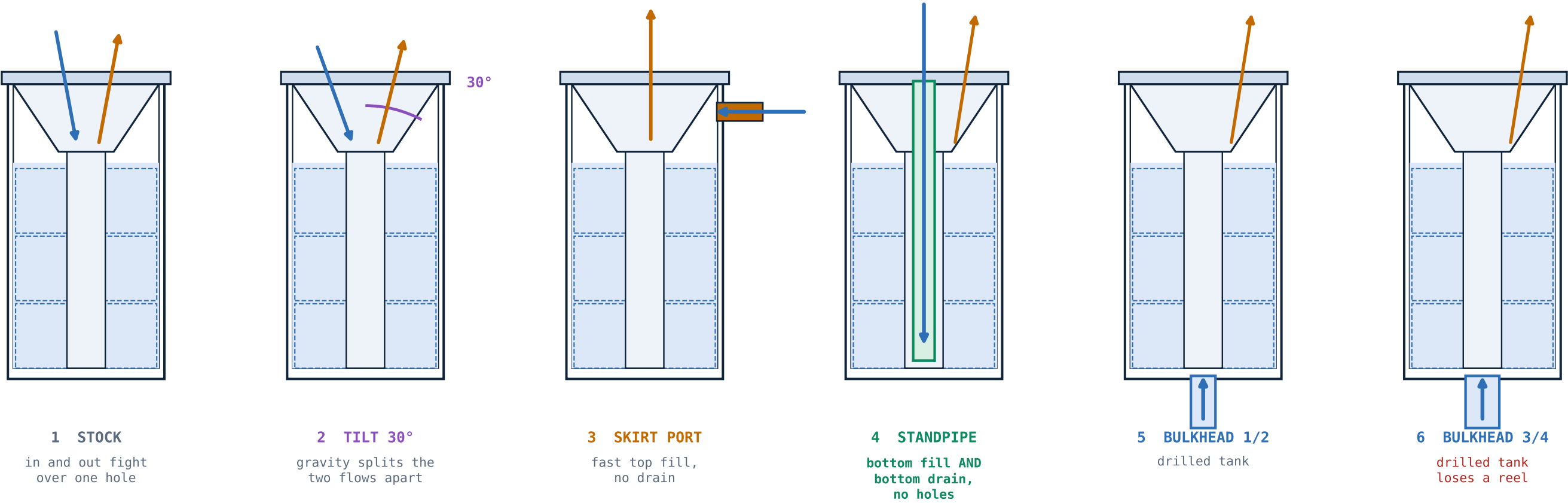
Development starts when the first film meets developer, so a 20 s fill means the bottom reel develops 20 s longer than the top one.

at 9 min	20 s is 3.7 %	invisible
at 6 min	20 s is 5.6 %	borderline
at 3 min	20 s is 11 %	a visible density gradient

Below about 4 minutes, no pouring technique is good enough. That threshold should decide whether you build any of this.

The six physical routes

where the liquid goes in, and where the air it displaces comes out



READING THE MAP

Blue arrows are chemistry, orange arrows are the air it displaces. A route is FAST exactly when the two arrows do not share a hole, and it is EVEN when the blue arrow enters at the bottom, so the liquid front rises through the reels pushing the air ahead of it. Routes 4, 5 and 6 do both. Route 3 separates the flows but still pours onto the top reel. Routes 1 and 2 do neither – route 2 simply makes the shared hole behave better by tipping it.

ONLY TWO ROUTES DRAIN FROM THE BOTTOM

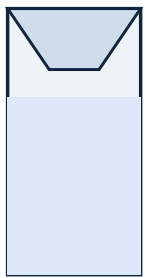
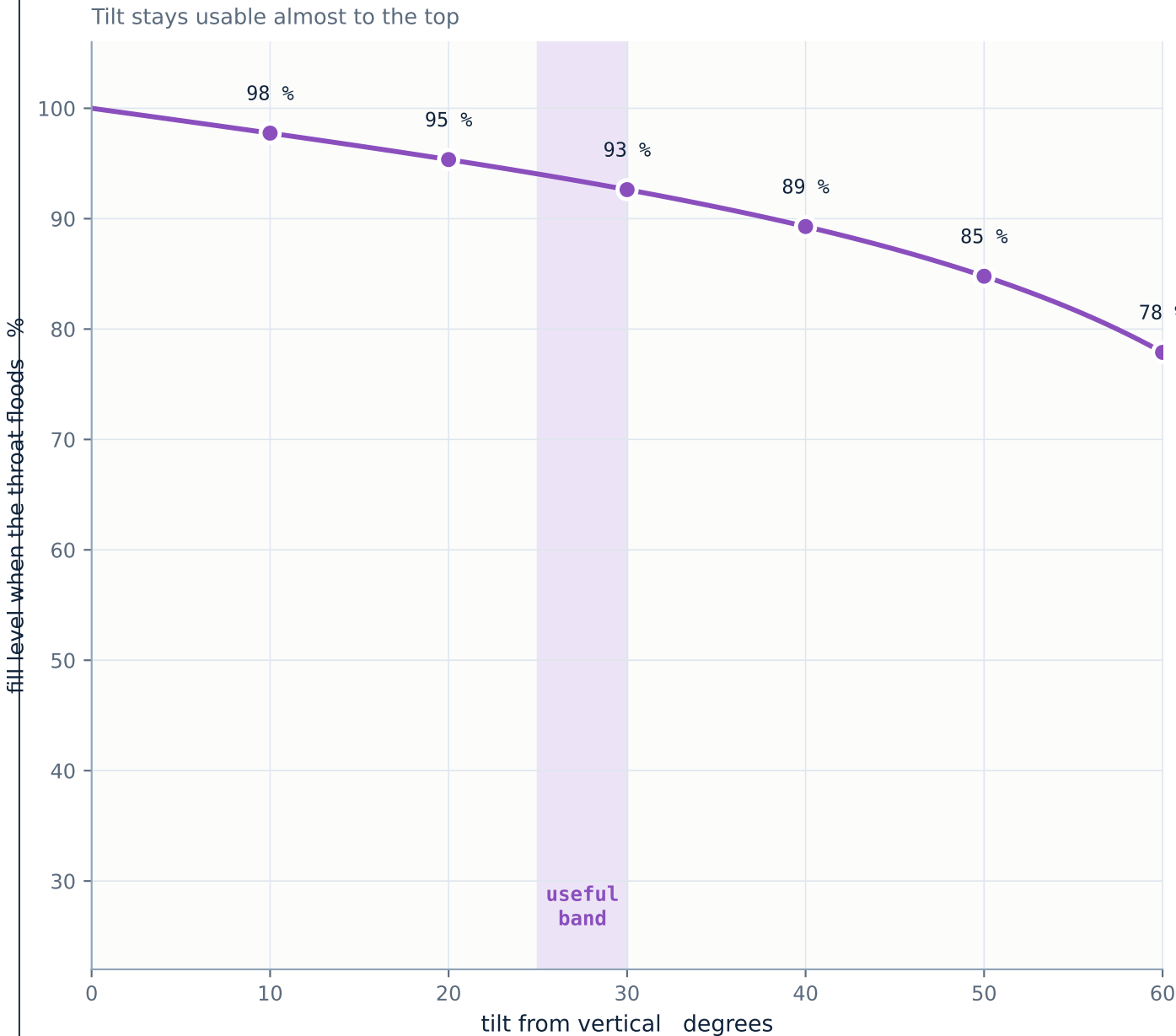
Route 4, and routes 5/6. Everything else drains by turning the tank upside down, because the reels leave only ~1.5 mm of radial clearance – no dip tube can run down the wall.

ROUTE 3 IS A HALF-MEASURE

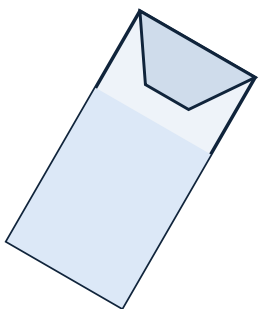
It buys the full fill speed and none of the drain speed or evenness. Worth it only if you refuse to touch the centre column. Sheet 7.

Tilting to fill — how far, and how far it gets you

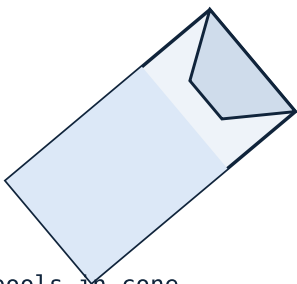
the throat floods from the inside only when the tank is nearly full



0° glugs



30° smooth



50° pools in cone

WHY TILTING WORKS

Upright, the throat is one vertical pipe carrying liquid down and air up at the same time. The phases fight, the flow goes to slug flow, and it burps.

Tilt it and gravity stratifies them: liquid runs down the low side of the throat, air escapes along the high side. You have made two channels out of one without touching the tank – in flooding terms, you have raised C.

Effect in practice: roughly 20 s → 10 s. About 2×, which matches the 1.5–2.5× that inclining a counter-current tube is generally worth.

WHY 25–30° AND NOT MORE

Two limits close in from opposite sides. BELOW ~20° stratification is incomplete and the flow still slugs intermittently. ABOVE ~35° chemistry pools in the funnel cone instead of draining through the throat, and the cone becomes the restriction instead.

What does NOT stop you is flooding from the inside. With the tank tilted θ the liquid climbs the low wall and reaches the throat lip when

$$z_{\text{mean}} + r_{\text{throat}} \cdot \tan\theta > 141 \text{ mm}$$

At 30° that is 130.6 mm – 93 % full, so you can hold the tilt through almost the whole pour and only bring the tank upright for the last splash.

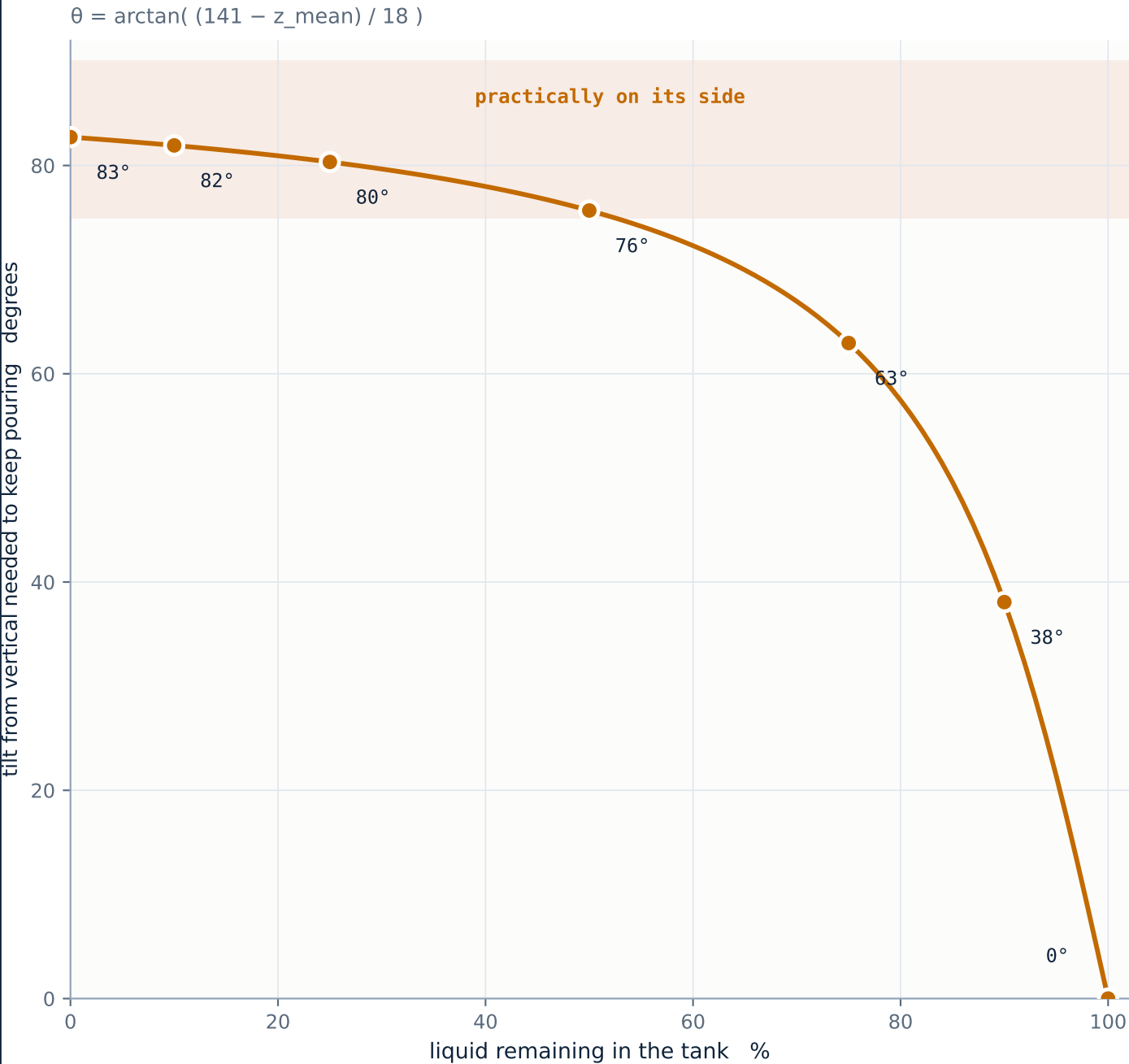
THE TECHNIQUE, WRITTEN OUT

- 1 Start at 25–30°, throat low-side down, jug spout 20–30 mm clear of the opening – never sealing it.
- 2 Pour steadily against the low wall of the cone.
- 3 At about 90 % – you will feel it slow – bring the tank upright and let the last 100 ml settle in.
- 4 Cap, then start the timer at the HALFWAY point of the pour, not the end. That halves the residual error for nothing.

Free, reversible, and it works on every tank in the range.

Tilting to drain — the geometry says no

you cannot pour this tank empty without very nearly inverting it



THE DERIVATION

Tilt the tank by θ . The free surface stays horizontal, so in the tank’s own frame it is a plane inclined at θ and the liquid climbs the low wall by $r \cdot \tan\theta$. It can leave only over the throat lip, 141 mm up the axis and 18 mm out from it, so it pours once the surface covers that lip:

	z_{mean}	$+ 18 \cdot \tan\theta$	> 141	
	θ	$> \arctan((141 - z_{\text{mean}}) / 18)$		
full	$z=141$	$\theta = 0^\circ$	spills at once	
90 % left	$z=127$	$\theta = 38^\circ$		
75 % left	$z=106$	$\theta = 63^\circ$		
50 % left	$z= 71$	$\theta = 76^\circ$		
25 % left	$z= 35$	$\theta = 80^\circ$		
bone dry	$z= 0$	$\theta = 83^\circ$		

The 18 mm throat radius is the villain. Even if the whole $\varnothing 95$ bore could discharge, the last drop would still need **AND THE CURVE UNDERSTATES IT**

That model is an open cylinder. The real tank adds three things it does not capture:

- the funnel cone is in the way and traps a pocket on the high side;
- surface tension holds 20–40 ml in the reel spirals at any angle – only a shake clears it;
- air has to come back IN as liquid leaves, so the counter-current fight from sheet 2 applies in reverse and it glugs the whole way down.

This is why Paterson’s own instruction is to invert. It is not laziness – it is the only angle that works.

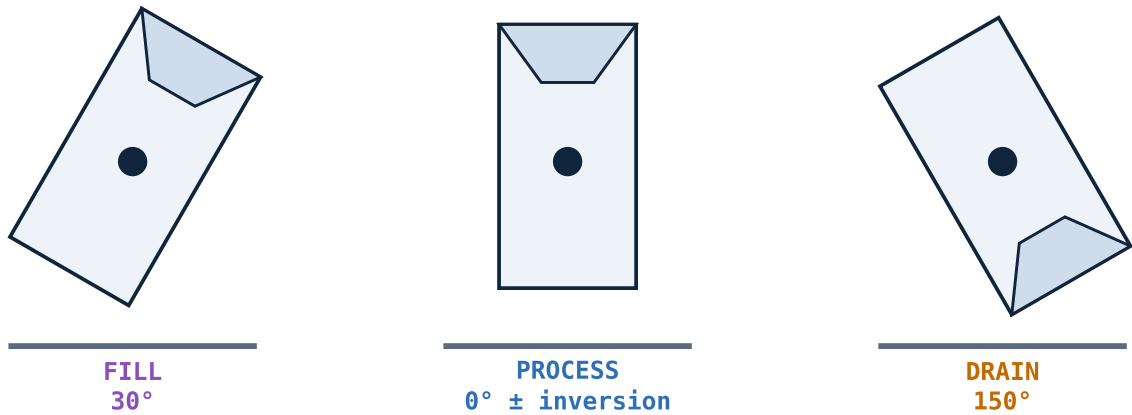
CONCLUSION ON TILT

Tilt is a FILLING tool. It roughly halves the fill time and costs nothing.

It is not a draining tool. Anything short of 80° leaves a quarter of your chemistry in the tank, and a partial tilt that dribbles for 30 s is worse for evenness than a clean 15 s inversion. For a fast drain you need a bottom path – route 4, 5 or 6.

Doing the tilting automatically

a three-position cradle – and why it is really an agitation machine



pivot through the tank axis at its loaded centre of mass
→ near-zero torque; a hobby servo is enough

SPECIFICATION

PIVOT	through the tank axis at 100 mm up – the loaded centre of mass. Balanced, so the motor only fights friction.
TORQUE	1.5 kg at 15 mm worst-case offset $\approx$ 0.22 N·m. A 20 kg·cm hobby servo is four times that.
STOPS	30° fill · 0° process · 150–180° drain.
ROTATION	continuous 180° for inversion agitation, one or two inversions per 10 s with a dwell at each end.
CRADLE	two printed saddles on $\varnothing$ 102, one M6 rod, a strap over the lid. The lid MUST be strapped – an EVA cap will let go at 180° under load.
CONTROL	three-position cam, or a servo with three preset angles and a cheap microcontroller.

BE HONEST ABOUT WHAT IT BUYS

FILLING	30° held perfectly still is worth maybe a second over doing it by hand. Not the reason to build it.
DRAINING	A motorised 150° dump genuinely beats hand inversion – repeatable, and it frees both hands. About 12 s, every time.
AGITATION	This is the real prize. An identical inversion pattern every run, every roll, forever. The largest source of run-to-run variation in hand processing is agitation, not fill time.
TEMPERATURE	The cradle can sit in a water bath. Add a recirculating pump and you have built a Jobo.

HOW IT COMBINES WITH THE OTHER ROUTES

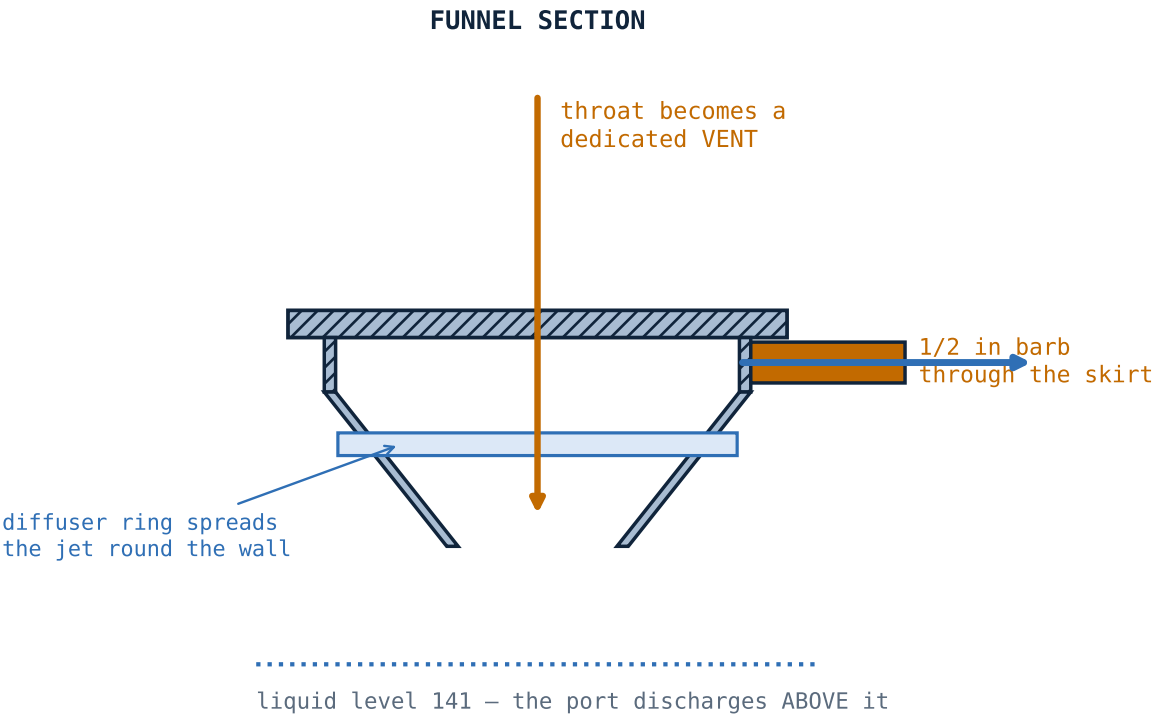
A tilt cradle and a bottom path are not alternatives – they solve different halves of the problem, and the best machine has both.

cradle alone	10 s fill · 12 s drain · perfect agitation · no plumbing and no holes
column standpipe alone	2.5 s fill · 9.7 s drain · hand agitation · one printed part
both	2.5 s fill · 9.7 s drain · perfect agitation – but the feed hose must release before the cradle inverts, which is exactly what the quick-disconnect on sheet 12 is for.

If you build one thing, build the standpipe. If you build two, add the cradle – for the agitation, not for the tilting.

Plumbing through the funnel — the skirt port

drill the cheap universal spare, not the tank



VERDICT ON THE SKIRT PORT

Half the benefit for a quarter of the nerve. Take it if you will not modify the centre column and you only care about fill speed — for instance if your problem is a 3-minute push and your drain already overlaps a rinse.

Otherwise go straight to the standpipe on the next sheet. It costs the same effort and does both jobs.

THE IDEA

The funnel and lid are a universal spare — SPTP110, sold for about the price of a roll of film and fitting every tank in the range. So modify that instead of the tank. Buy two: one stock, one drilled. Swap back in thirty seconds; sell the tank on unmarked.

Fit a 1/2 in barb through the SKIRT — the cylindrical part that seals into the $\varnothing 95$ bore — so it discharges below the light trap, straight into the tank. The throat above then carries nothing but air.

That is the whole trick: you have not made the hole bigger, you have given the air its own hole.

WHAT IT DELIVERS

fill	2.7 s	identical to a 1/2 in bottom bulkhead
drain	15 s	unchanged, still by inversion
tank	untouched	

The fill figure is real: once the vent is separate, the port is the only restriction left, so it obeys the same gravity equation as any other 1/2 in line.

THE TWO CATCHES

- 1 IT IS STILL A TOP FILL. Chemistry lands on the top reel and works down. You get the speed but not the even rising front a bottom feed gives, and a bare jet will hammer the top film — hence the diffuser ring.
- 2 IT DOES NOTHING FOR DRAINING. The reels leave only about 1.5 mm of radial clearance against the bore, so no dip tube can reach the floor down the wall.

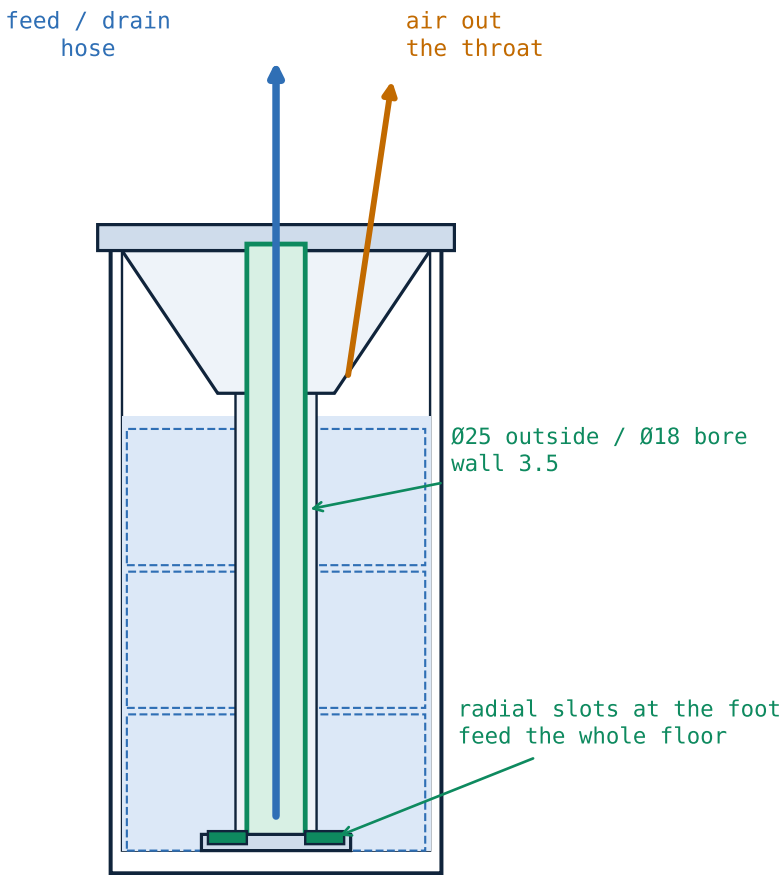
A hole in the skirt is a straight line to the film. Trap it exactly like a bulkhead: opaque hose, 300 mm, one 90°

A NOTE ON VENT SIZING

The vent must pass air at the same volumetric rate the liquid arrives — 370 ml/s at 0.6 m of head. Through the $\varnothing 36$ throat that is 0.36 m/s of air and a pressure drop well under 1 Pa. Air is a thousand times lighter than developer; venting is never the limit once it has a path of its own. Leave the throat stock.

The centre-column standpipe — best route overall

bottom fill and bottom drain, and not one hole in the tank



WHAT YOU BUILD

One printed part: a replacement centre column whose external geometry matches SPTP116 exactly, so the reels and funnel cannot tell the difference – but with a Ø18 mm bore running its full length and four radial slots at the foot. The top takes a push-fit hose tail instead of, or as well as, the agitator-rod socket, and a hose drops down through the funnel throat and clips on.

PETG rather than PLA. Four perimeters, no sparse infill in the wall, and soak-test it for a week before it ever sees film – a delaminated column is a light leak.

WHY IT BEATS DRILLING THE TANK

	Standpipe	1/2 in bulkhead	Winner
fill, 0.6 m head	2.5 s	2.7 s	standpipe
drain, free	9.7 s	10.3 s	standpipe
holes in the tank	none	one	standpipe
reversible	yes	no	standpipe
light risk	low	medium	standpipe
keeps 3 reels	yes	yes	tie
can still invert	unclip	QD needed	standpipe

THE ONE REAL RISK

A printed column is structural – the reels and funnel lock to it. If it cracks mid-process the tank is no longer light-tight. Print two, soak-test both, and keep the stock column.

THE HYDRAULICS

A Ø18 bore, but a long one – 190 mm of column plus the hose above it – so the discharge coefficient falls from 0.55 to about 0.45 for pipe friction and two bends.

fill t = 2.5 s from 0.6 m of head
drain t = 9.7 s free discharge

Even carrying that penalty it still edges a 1/2 in bulkhead, because 18 mm of bore beats 15.8 mm by more than the friction costs.

Do not go bigger. The column is Ø25 outside and the reels hang on it; a Ø20 bore leaves a 2.5 mm wall that will flex when three loaded reels are pushed down it.

LIGHT-TIGHTNESS — WHY THIS IS THE SAFER OPTION

The feed path is a black tube inside a black tube inside a light-trapped funnel. Light entering the throat must turn a corner at the hose, travel 190 mm down a matt bore, then turn 90° again at the foot slots before it could reach any film – which is wound on reels facing outward, away from the column.

Compare a bulkhead: a straight hole through the floor into the chamber the film is sitting in. Both can be made safe. Only one is safe by accident.

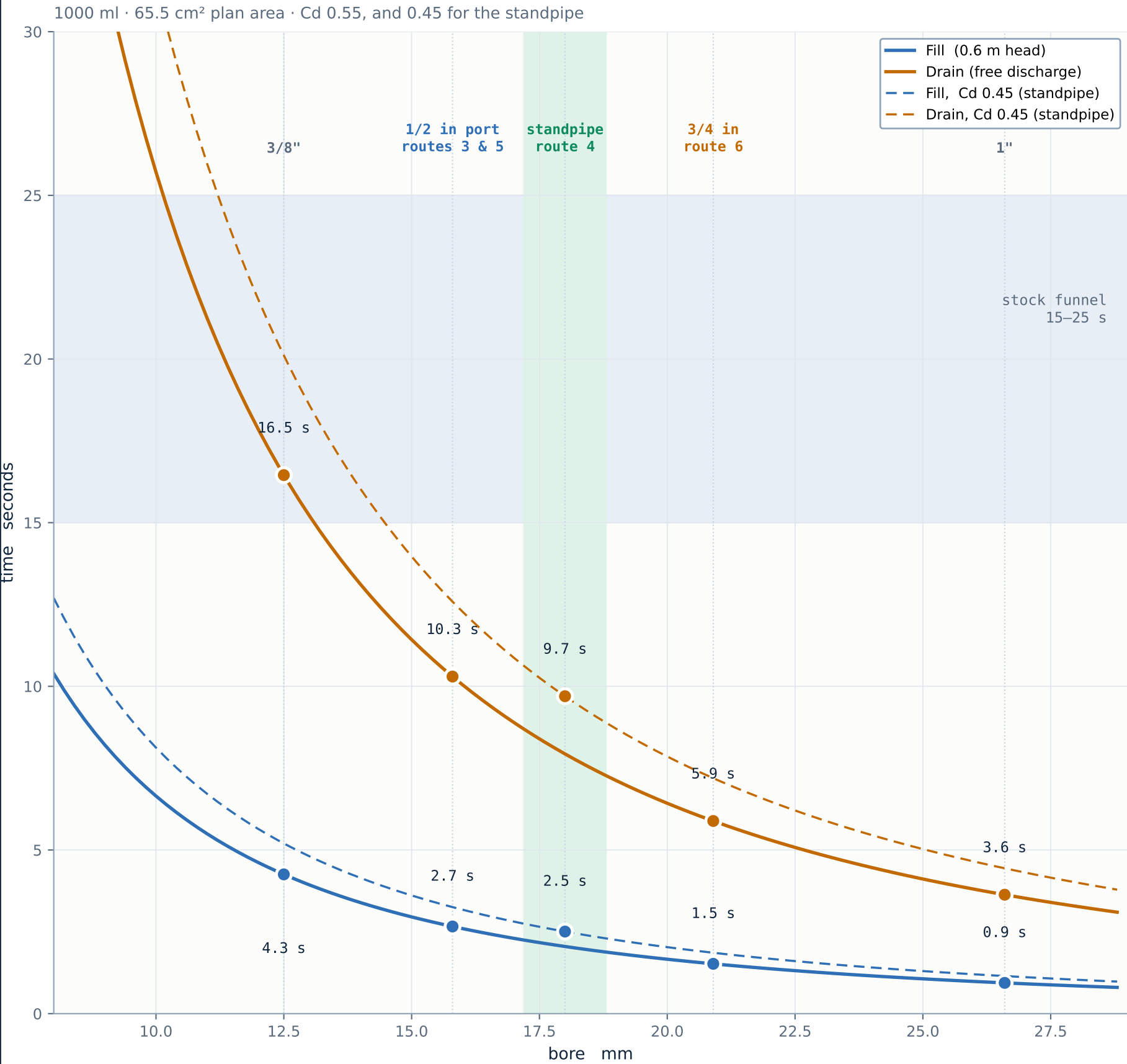
OPERATING IT

Fill: clip on, open the clamp, count to three, unclip, cap the column head, agitate as normal.

Drain: clip on, drop the hose end below the bench, let it siphon. No pump, no power.

Sizing any bottom path

fill and drain time against bore, with every route marked



BOTH CURVES GO AS 1/d²

3/8 → 1/2 in	fill 4.3 → 2.7 s	saves 1.6 s
1/2 → 3/4 in	fill 2.7 → 1.5 s	saves 1.2 s
3/4 → 1 in	fill 1.5 → 0.9 s	saves 0.6 s

On filling, 3/4 in buys about a second over 1/2 in – less than your reaction time on a timer. On draining it is worth more, 5.9 s against 10.3 s, but sheet 10 shows what that costs in reels.

SIZING FROM FIRST PRINCIPLES

Target: fill in under 5 % of the development time.

9 min development	→	27 s allowed
6 min	→	18 s
3 min push	→	9 s
90 s E6 first dev	→	4.5 s

Everything from 3/8 in upward clears the 3-minute bar. Only 1/2 in and above clears the 90-second bar. Since the hydraulic choice is therefore nearly free, make it on mechanical grounds instead – which is sheet 10.

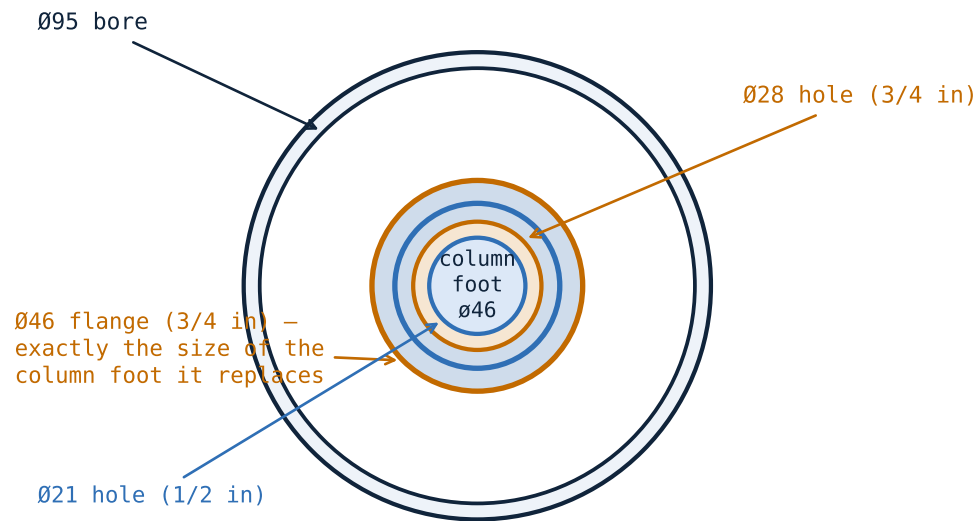
WHERE THE CURVES STOP HELPING

Below about 2 s the fill is no longer limited by the bore at all. It is limited by how fast you can open a valve, and by the air still leaving through the funnel.

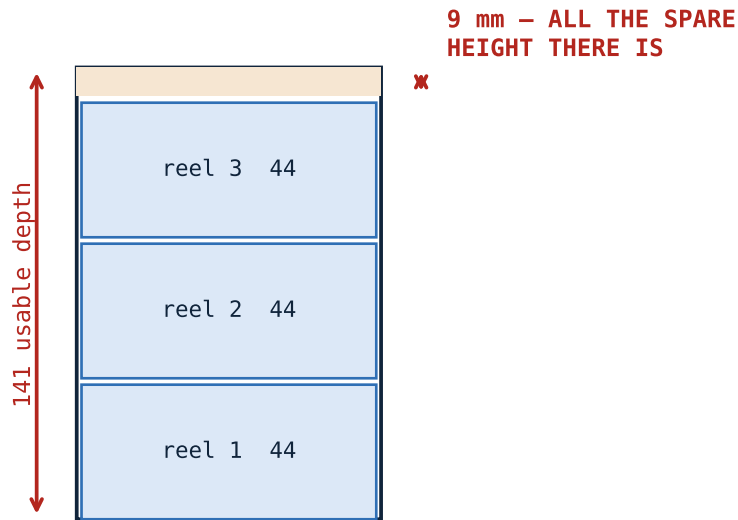
Past ø20 you are engineering for a second you cannot measure. Stop at the standpipe.

If you do drill it — will 3/4 in fit?

the floor plan says yes, the headroom says only just



FLOOR PLAN — CENTRE MOUNTED



HEIGHT STACK-UP

1 THE FLOOR PLAN — PASSES, BUT ONLY AT THE CENTRE

Off-centre is impossible at any size. To clear the Ø46 column foot the port centre must sit at $r \geq 38$ mm; to keep its flange inside the Ø95 bore it must sit at $r \leq 32.5$ mm. No overlap, even for 3/8 in.

Centre mounted, a 3/4 in flange (Ø46) sits inside the bore with 24 mm clearance all round. So the floor is not the constraint — but centring means the column foot has to go whatever bore you pick, which is most of the work of building a standpipe anyway.

2 THE HEADROOM — THIS IS WHAT DECIDES IT

usable depth to the funnel	141 mm
three reels at 44 mm pitch	132 mm
budget for everything inside	9 mm

1/2 in low-profile nut	6 mm, legs straddle it
3/4 in low-profile nut	9–10 mm, they cannot

Exceed 9 mm and the top reel fouls the funnel: the lid will not lock, and a 3-reel tank has quietly become a 2-reel tank.

VERDICT ON DRILLING

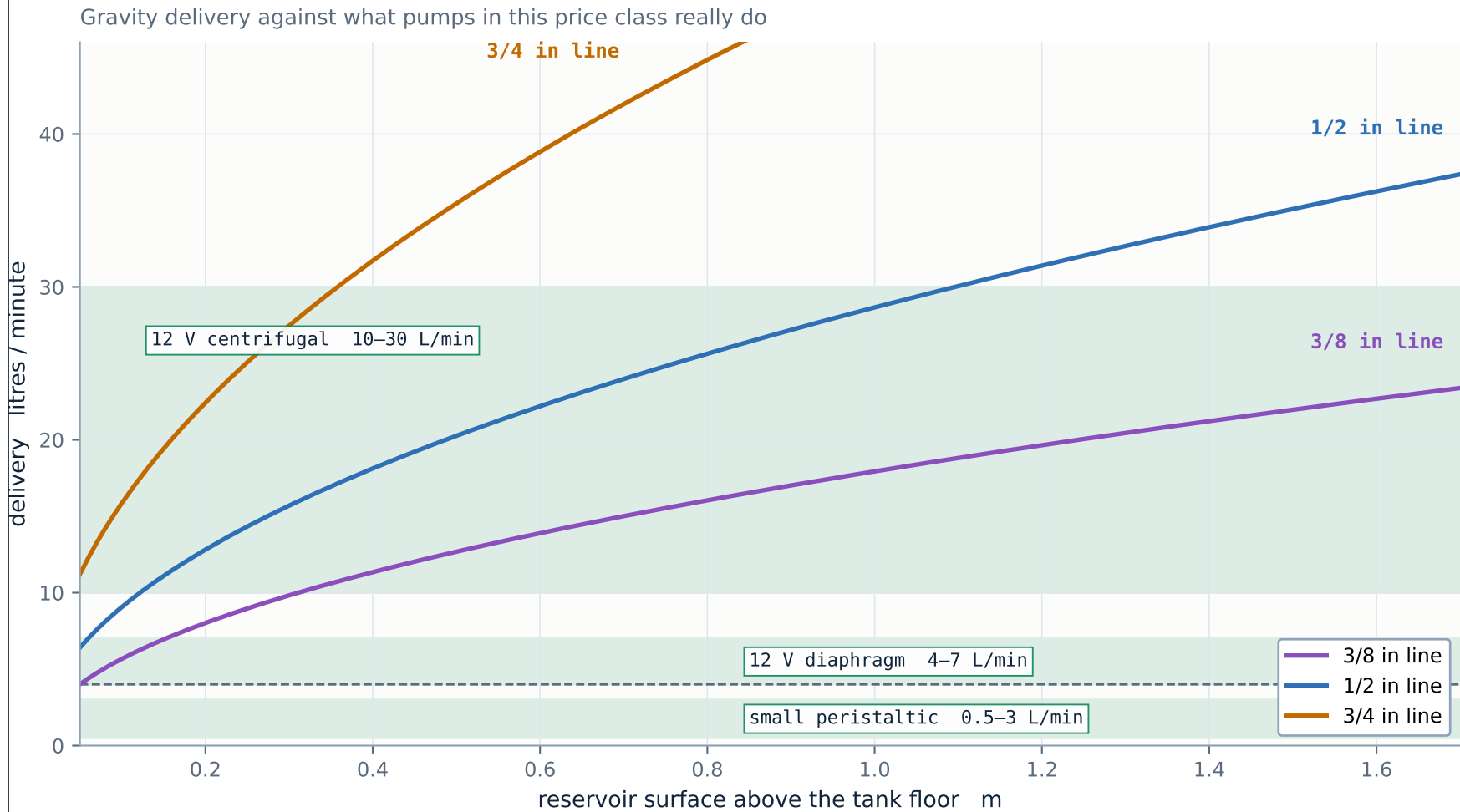
3/4 in is not too wide for the floor — it is too tall for the ceiling. Fitted dead centre it works, fills in 1.5 s, drains in 5.9 s, and will almost certainly cost you the third reel. If you routinely run two rolls, that is a fair trade.

If you want three rolls and you are set on drilling, fit 1/2 in: 2.7 s and 10.3 s, full capacity.

But note what sheet 8 showed. Both sizes require replacing the column foot, which is the same work as printing a standpipe — and the standpipe then gives 2.5 s and 9.7 s with no hole at all. The only remaining reason to drill is if you specifically want 3/4 in drain speed and will trade a reel for four seconds.

Pump or gravity?

gravity wins on speed; the pump wins at everything it is actually for



THE HEADLINE

A shelf 600 mm above the bench feeding a 1/2 in line delivers 370 ml/s – 22 litres a minute.

To match that you need a 22 L/min pump that is also fixer-proof. A cheap diaphragm pump gives 4–7; a small peristaltic gives 0.5–3, slower than the stock funnel.

Even 200 mm of head beats every pump here bar a large centrifugal – which is rated at zero head anyway.

SO USE GRAVITY FOR FILL AND DRAIN

- | | |
|-------|--|
| fill | reservoir on a shelf 0.4–0.8 m up, gate clamp or ball valve at the tank. No power, no priming, nothing extra to flush. |
| drain | tank on the bench, waste on the floor. 0.3 m of drop is plenty. |

Volume is set by measuring into the reservoir, not by timing a pump – one less thing to get wrong.

AND USE A PUMP FOR WHAT IT IS GOOD AT

- | | |
|---------------|--|
| RECIRCULATION | one turnover per 15 s is only 4 L/min, and it gives continuous, perfectly repeatable agitation with no inversion at all. The real reason to own one. |
| LAST RESIDUE | gravity leaves 20–40 ml in the spirals; a peristaltic pulls it out, gravity will not. |
| NO SHELF | accept 8–15 s instead of 2.5 s. |
| TEMPERING | circulate through a coil in a water bath. |

CARRY-OVER – THE HIDDEN COST

Every metre of hose and every pump head holds chemistry the next step inherits. A 1/2 in line 0.5 m long holds 98 ml – a tenth of your tank volume. Fixer carried into developer is the one that ruins film.

- one dedicated line per chemical, or
- a water flush between every step, or
- quick-disconnects and one hose set per chemical.

Gravity has the same problem and the same answers, with one fewer wetted part to flush.

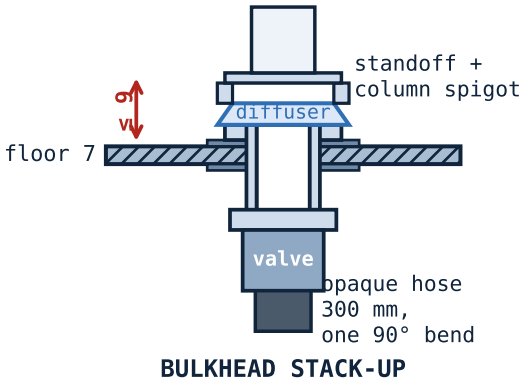
CHOOSING A PUMP, IF YOU BUY ONE

- PERISTALTIC** Only the tube touches chemistry, so compatibility is a tubing question and nothing else. Self-priming, safe to run dry, reverses for fill and drain with one unit, meters volume by revolution count. Slow. Norprene or platinum-cure silicone.
- DIAPHRAGM** The practical middle. Demand a Santoprene or EPDM diaphragm and a polypropylene head. Self-priming, dry-run safe, 4–7 L/min, and cheap.
- CENTRIFUGAL** Fast and cheap, but not self-priming and destroyed by running dry. Only worth it if it stays flooded.
- NEVER** Brass, bronze or aluminium wetted parts – fixer attacks them and dissolved copper fogs film. Gear pumps. Anything you cannot flush completely between developer and fixer.

Decide, then build

the matrix, the stack-up, and the test you must not skip

If this is you	Build this	Cost	Fill / drain s
Shortest development ≥ 6 min	2 tilt technique	free	10 / 15
≥ 4 min, no tools	2 tilt technique	free	10 / 15
< 4 min, will print a part	4 column standpipe	low	2.5 / 9.7
< 4 min, will not print	3 funnel skirt port	low	2.7 / 15
Want the fastest drain	6 3/4 in bulkhead	med	1.5 / 5.9
Want repeatability	8 tilt cradle + 4	med	2.5 / 9.7
Want it hands-off	7 recirc pump + 4	high	2.5 / 9.7



MATERIALS

- USE

PP or PVDF bulkheads

EPDM or silicone gaskets

316 stainless if metal

black EPDM or silicone hose

PETG for printed parts
- NEVER

brass or bronze – fixer eats it, copper fogs film

aluminium – developer eats it

natural rubber – swells

clear hose of any colour
- DO NOT CUT A THREAD IN THE TANK.

7 mm of polystyrene splits on a tapered NPT thread – if not on assembly then on the third thermal cycle. Through-hole, gasket both sides, nut takes the load.

DRILLING HIPS

- Back the floor with a clamped scrap block.
- Step drill, never a hole saw – it grabs and splits.
- Low speed, no pressure.
- Chamfer both faces 0.5 mm to kill crack initiation.
- Nut hand tight plus a quarter turn; HIPS creeps.

COMMISSIONING, IN ORDER

- Water only. Check for leaks at temperature, twice.
- Time a 1000 ml fill and a full drain; those numbers are now part of your process.
- Light-trap test, right. Then one roll of cheap film.

LIGHT-TRAP TEST – THE ONE YOU MUST NOT SKIP

Any new opening is a light leak until proven otherwise. Light travels in straight lines, so 300 mm of opaque hose with one 90° bend defeats it; a closed valve defeats it absolutely.

- Load a strip of the fastest film you use.
- Assemble fully, port open, hose connected.
- Stand it in the brightest light you have for ten times your longest process time.
- Develop it beside an unexposed control from that roll.
- Any density difference at all – the trap has failed. Lengthen the hose or add a bend. Do not rationalise it.

THE TRADE-OFF TO SETTLE BEFORE YOU BUILD ANYTHING

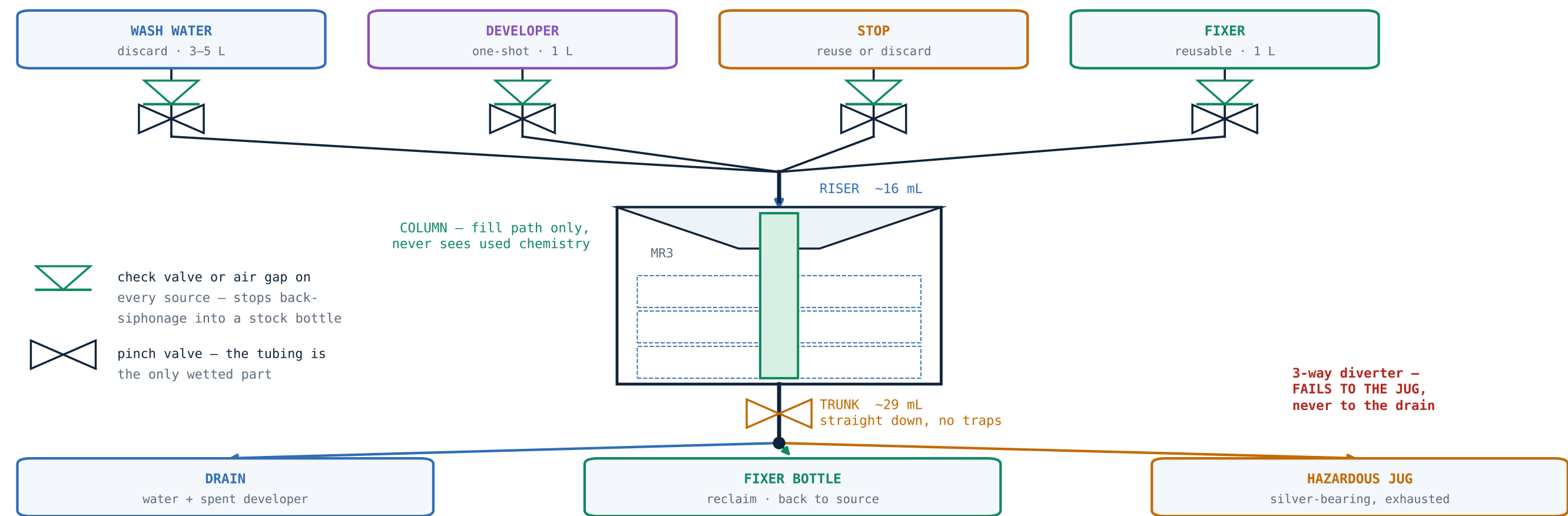
A hose attached to the tank cannot be inverted, and inversion is the agitation method the tank was designed around. Three ways out:

- Rotary only
- Quick-disconnect
- Recirculation
- Use the agitator rod. Simplest, slightly more prone to edge effects. Re-test your times.
- Self-sealing QD at the port. Fill, disconnect, invert exactly as now, reconnect to drain. Keeps every existing time.
- Pump continuously through the loop. Most even and most repeatable, and the largest build.

The quick-disconnect is the right first build whichever route you pick: reversible, preserves your development times, costs one fitting.

Fluid architecture

four sources, one tank, three destinations – and nothing shared that matters



THE TWO RULES THAT DO MOST OF THE WORK

- 1 EVERYTHING BELOW THE TANK SLOPES DOWN to its destination. No horizontal runs, no U-traps, no loops. Then every path empties itself after every step and carry-over is a wall film, not a slug. Worth about 40x; sheet 14 does the arithmetic.
- 2 ROUTE THE WASH THROUGH THE SHARED PATH. By the end of a wash the trunk has had 3-5 L of water through it. The process you already run IS the flush.

WHY FILL AND DRAIN ARE SEPARATE PATHS

Fill enters over the top, down the printed column, out at the floor. Drain leaves through a bulkhead in the floor, straight down. They share nothing but the tank.

CORRECTION TO SHEET 8: a standpipe used for BOTH fill and drain has to siphon up over the column head at 190 mm – above the 141 mm liquid surface. It will not self-prime. It works only if you drain immediately after filling, while the line is still full. For unattended use, drill the floor for the drain or pump the drain leg

WHY THIS SHAPE AND NOT A VALVE MATRIX

A full N×M matrix lets any source reach any destination. That sounds flexible and is actually a liability: one stuck valve can put fixer into the developer bottle.

This layout is deliberately poorer. Sources only ever go one way – into the tank. Destinations are only reachable from the tank. Nothing routes bottle-to-bottle, so no single failure can cross-contaminate a stock solution. The worst case is a ruined run, not a ruined litre.

IF YOU WILL NOT DRILL

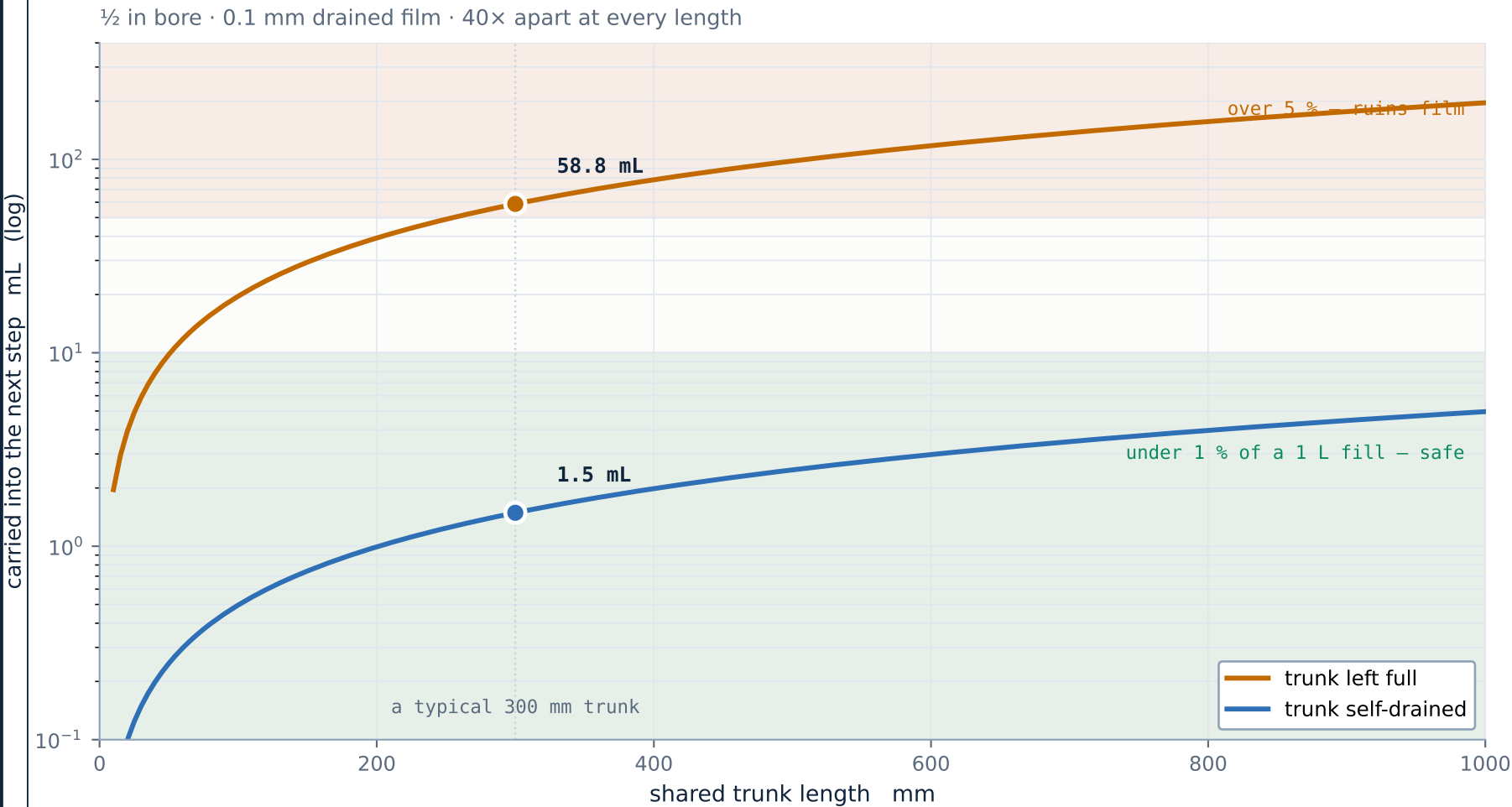
Run everything through the column and accept one high point at the column head. Then you need:

- an air-admittance valve at the apex. It breaks the siphon and lets both legs drain, and it doubles as the back-siphonage guard;
- the drain sequenced straight after the fill so the line stays primed, or a pump on the drain leg.

Costs one component and some sequencing discipline, and keeps the tank unmarked.

Carry-over budget

the one number that decides whether any of this works



THE COMPONENT BUDGET

path element	left full mL	self-drained mL
Column Ø18 × 190	48.3	1.1
Trunk ½ in × 300	58.8	1.5
Valve bodies	~5	~0.2
TOTAL	112	2.8
as % of a 1 L fill	11.2 %	0.28 %

AND THE PURGE IT IMPLIES

To flush a residue down to 1 mL of the old liquid, a well-mixed line needs $V = V_{\text{res}} \cdot \ln(V_{\text{res}} / 1 \text{ mL})$:

left full 112 mL → purge 528 mL
self-drained 2.8 mL → purge 2.9 mL

Half a litre per step, or three millilitres. That ratio is the whole argument for fixing the geometry first.

WHERE 0.1 mm COMES FROM

A low-viscosity aqueous solution draining off a vertical plastic wall leaves a film of roughly 0.05–0.15 mm. 0.1 mm is the middle of that band.

It scales with the square root of viscosity, so cold, syrupy stock leaves more and warm working-strength solution leaves less.

Call the self-drained figure 2–5 mL rather than 2.8 mL and nothing changes: still forty times better than leaving the line full, still under half a percent of the tank volume.

THE TWO WAYS CARRY-OVER HURTS – THEY ARE NOT THE SAME

POISONING Fixer into developer. A few mL per litre is enough to depress contrast and cause dichroic fog. Cumulative and irreversible. This is the one that scraps film.

DILUTION Wash water into developer. 112 mL of water in a 1 L fill is an 11 % dilution – an 11 % change in development activity, which reads as flat, thin negatives. Predictable, repeatable, and completely avoidable.

Self-draining geometry fixes both at once. Where it cannot – the developer always follows the wash – a purge to waste is the answer, and at 3 mL it is not worth automating around.

Process sequences

who follows whom, and the one step that needs a purge

BLACK & WHITE – one-shot developer, reusable fixer

#	Step	Liquid	Class	Vol	Source	Destination	Purge first?
1	Pre-soak (optional)	water	discard	1 L	wash line	drain	no
2	Developer	one-shot	discard	1 L	dev reservoir	drain	YES ~150 mL
3	Stop	water or acid	either	1 L	wash / stop bottle	drain / stop bottle	no
4	Fixer	reusable	reclaim	1 L	fixer bottle	fixer bottle	no
5	Wash	water	discard	3-5 L	wash line	drain	no — this IS the flush
6	Wetting agent	one-shot	discard	1 L	small reservoir	drain	no

C-41 – every bath reusable, 38 °C, and it breaks the easy rule

#	Step	Liquid	Class	Vol	Source	Destination	Purge first?
1	Pre-warm	water 38 °C	discard	1 L	tempered wash	drain	no
2	Colour developer	replenished	reclaim	1 L	dev bottle	dev bottle	YES → replenisher
3	Bleach	reusable	reclaim	1 L	bleach bottle	bleach bottle	no
4	Wash	water	discard	2 L	wash line	drain	no
5	Fixer	reusable	reclaim	1 L	fixer bottle	fixer bottle	no
6	Wash	water	discard	3 L	wash line	drain	no
7	Stabiliser	reusable	reclaim	1 L	stab bottle	stab bottle	no

THE FREE WIN – CONTAMINATION ONLY MATTERS ONE WAY

In both sequences each bath is preceded by something harmless to it:

- stop ← developer neutralised anyway
- fixer ← stop acid into acid fixer
- wash ← fixer irrelevant
- developer ← wash DILUTION – the only harmful one

So exactly one step needs a purge, and in B&W it is the one-shot, where 150 mL to waste costs pennies.

WHERE C-41 IS HARDER

The colour developer is reusable AND follows a wash, so its purge cannot go to waste without losing stock. Send it to the replenisher jug and replenish on volume.

Bleach carried back into the developer is the serious one – it oxidises the developing agent. Never sequence bleach before developer, and if a blix replaces steps 3 and 5, put a wash either side of it.

All lines and bottles must hold 38 °C ± 0.3.